

INTRODUCTION TO ENGINEERING · LAB 2

Systems Thinking I

Boundaries, inputs & outputs, stakeholders, and operating context — applied through two games and a graded group challenge.

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Lab Agenda



Recap

Systems, boundaries, operating context & stakeholders from Lecture 2.

15–20 min



Game 1 — Reverse Engineer an Everyday System

Teams draw a system card and break it into elements, boundary, inputs & outputs.

15 min



Game 2 — Stakeholder Detective

Same cards, new lens: map Uses It / Runs It / Affected By It.

15 min



Group Work — Applied Systems Challenge

Scenario cards; graded on all six systems-thinking steps.

30 min · evaluated

QUICK RECAP

Systems, Boundaries & Operating Context

A **system** is elements + processes working together with purpose. Its **boundary** is a choice the engineer makes, not a fact about the world — and the **operating context** is everything just outside it that still constrains behavior.



Regulations & codes

Health codes, safety standards, building codes.



Time & schedule

Opening hours, semester calendar, deadlines.



Existing infrastructure

What's already built that the system must work within.



Budget & resources

What funding, staff, or materials are actually available.

Remember: the boundary is drawn, not discovered.

Two engineers can draw the boundary of the same system in different places — both can be valid, as long as the choice is justified and consistent.

QUICK RECAP

Who Are the Stakeholders?

A stakeholder is anyone who uses the system, runs it, or is affected by it — whether or not they sit inside the boundary.



Uses It

The people who directly interact with the system day to day.



Runs It

The people who operate, maintain, or keep the system running.



Affected By It

People touched by the system's effects — often from outside the boundary.

WHY IT MATTERS

A stakeholder map is a simple, useful first draft — sort names into the three columns above, then look for the ones who disagree with each other. Today's games and group work all build this map for a real system.

Today's goal: practice naming a system, drawing its boundary, tracing inputs/outputs, mapping stakeholders, proposing a solution, and noting the operating context — on real scenarios.

GAME 1 · 15 MINUTES

Reverse Engineer an Everyday System

Take a system apart to see how systems thinking really works.



How to Play: Reverse Engineer an Everyday System



1. Draw a card

Each team draws one or more card describing an everyday system (e.g. a vending machine, a traffic light, a coffee shop).



2. Break it apart

Identify the system's key elements and the processes connecting them.



3. Draw the boundary

Sketch a boundary line and list at least 3 inputs and 3 outputs crossing it.



4. Report back

One spokesperson explains the system in under 60 seconds when time is called.

I will hand each team a printed card with a different everyday system — no two teams get the same one.

Example Card: The Traffic Light Intersection

INPUTS

Vehicle & pedestrian sensors →

Electricity supply →

Timing schedule / signal plan →

System boundary



**Traffic Light
Intersection**

OUTPUTS

→ Red / yellow / green signals

→ Traffic flow data

→ Pedestrian “walk” signal

Operating context: *city traffic code · rush-hour schedule · existing road layout · municipal maintenance budget*



GAME 2 · 15 MINUTES

Stakeholder Detective

Same systems, new question: who uses it, runs it, and is affected by it?

How to Play: Stakeholder Detective



1. Keep your card

Teams reuse the same everyday-system card from Game 1 (or swap with another team).



2. Investigate

List every stakeholder you can think of — direct users, operators, and people affected from outside.



3. Sort them

Place each stakeholder into Uses It / Runs It / Affected By It on your worksheet.



4. Spot the tension

Flag one pair of stakeholders whose interests conflict, and be ready to share it.

Good detectives find stakeholders outside the boundary too — not just the obvious users inside it.

Example Card: The University Makerspace (3D-Printing Lab)

A stakeholder is anyone who uses, runs, or is affected by the makerspace — whether or not they sit inside its boundary.



Uses It

Students printing course projects; club teams printing parts.



Runs It

Lab technicians, work-study student staff, procurement office.



Affected By It

Neighboring-room occupants (noise, fumes); building fire-safety officer.



The tension

Students want 24/7 open-access printing. The fire-safety officer wants printers supervised and powered off overnight. Procurement wants filament costs capped.

Who's right depends on which stakeholder's goals you weigh most — that's exactly the judgment call the group work asks you to make.

GROUP WORK · 30 MINUTES · EVALUATED

Applied Systems Challenge

Your team gets one real conflict scenario — work it end to end.



Your Task: Six Steps, One Scenario

Each team receives a different scenario card with a real conflict between stakeholders. Work through all six steps below and be ready to present.

1

Name the system

Give it a precise name, following the naming principles from Lecture 2.

2

Draw the boundary

Decide what's inside vs. outside, and justify the choice.

3

List inputs & outputs

At least 3 inputs and 3 outputs crossing the boundary.

4

Stakeholder analysis

Sort stakeholders into Uses It / Runs It / Affected By It.

5

Agreed solution

Propose one solution to the scenario's conflict — and say who it favors.

6

Operating context

Name the regulations, schedule, infrastructure, or budget constraints at play.

Graded live: I'll circulate with a checklist and score each team on all six steps before time is up.

Campus EV Charging

CAMPUS EV CHARGING



The university plans 8 EV chargers for campus parking. Students want more chargers, facilities staff worry about peak electricity demand, and nearby offices want parking access preserved. Analyze the charging system and suggest one solution.

Task: name system, draw boundary, list inputs/outputs, map stakeholders, propose solution, note operating context.

Campus EV Charging

8 chargers planned · students want more · facilities worry about peak demand · offices want parking preserved.

1 · System: Campus EV Charging Station Network

INPUTS

OUTPUTS



4 · Stakeholders

Uses It	Runs It	Affected
Students & staff driving EVs	Facilities & maintenance team	Nearby offices; gas-car parkers

5 · Agreed solution:
Install 8 chargers across the two least-used lots; cap charging to 2 hours during peak (9am–5pm) to manage grid demand; reserve 2 EV-only bays exempt from general parking to protect office access.

6 · Operating context: campus electrical grid capacity · university parking policy · 8-charger installation budget · peak-hour class schedule (9am–5pm)

This is a model answer — your team's scenario is different, and a different (well-justified) solution can score just as well.

THANK YOU

Questions?

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